The Squat

The squat has been the most important, yet most poorly understood, exercise in the training arsenal for a very long time. The full-range-of-motion exercise known as the squat is the single most useful exercise in the weight room, and our most valuable tool for building strength, power, and size.

The squat is literally the only exercise in the entire repertoire of weighted human movement that allows the direct training of the complex movement pattern known as *hip drive* – the active recruitment of the muscles of the posterior chain. The term *posterior chain* refers to the muscles that produce hip extension – the straightening out of the hip joint from its flexed (or bent) position in the bottom of the squat. These muscle groups – also referred to as the *hip extensors* – are the hamstrings, the glutes, and the adductors (groin muscles). Because these important muscles contribute to jumping, pulling, pushing, and anything else involving the lower body, we want them strong. The best way to get them strong is to squat, and if you are to squat correctly, you must use hip drive, which is best thought of as a shoving-up of the sacral area of the lower back, the area right above your butt. Every time you use this motion to propel yourself out of the bottom of the squat, you train the muscles in the posterior chain.

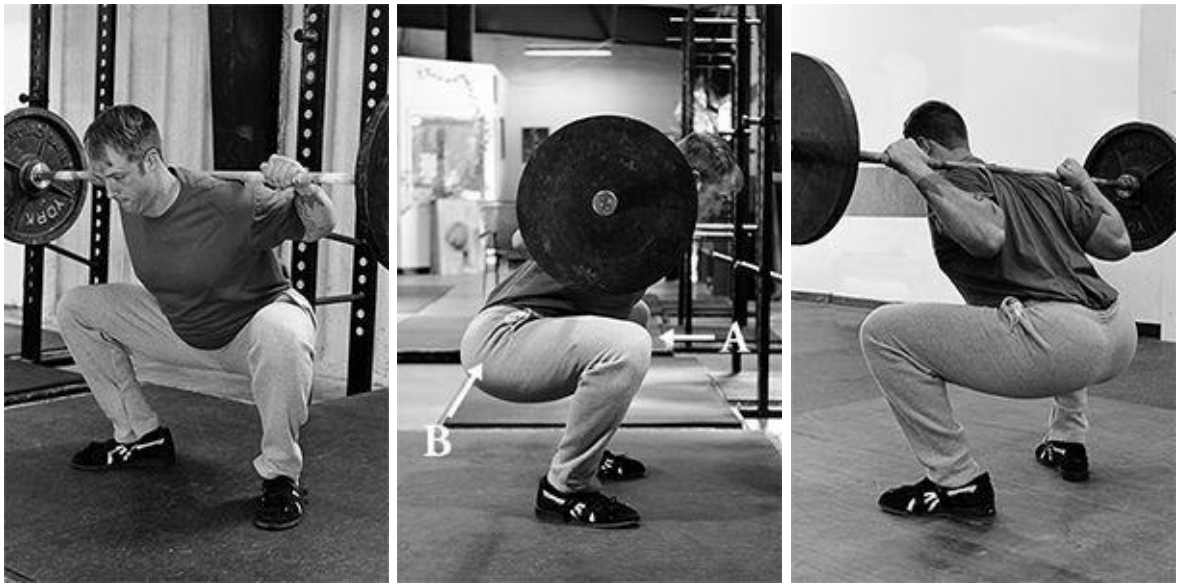


Figure 2-1. Three views of the squat. *Profile view*, Depth landmarks for the full squat. The top of the patella (A) and the hip joint, as identified by the apex in the crease of the shorts (B). The B side of the plane formed by these two points must drop below parallel with the ground.

All styles of squatting tend to make the quads sore, more so than any of the other muscles in the movement. This soreness occurs because the quads are the only knee extensor group, while the hip extensors consist of three muscle groups (hamstrings, glutes, adductors). They comprise more potential muscle mass to spread the work across – if they are trained correctly. Given this anatomical situation, we want to squat in a way that maximizes the use of all the muscle that can potentially be brought into the exercise and thus be strengthened by it. So we need a way to squat that involves the posterior muscle mass, making it operate up to its potential for contributing to strength and power. The low-bar back squat is that way.

Done correctly, the squat is the only exercise in the weight room that trains the recruitment of the entire posterior chain in a way that is progressively improvable. These are the things that make the squat the best exercise you can do with barbells and, by extension, the best strength exercise there is. The squat trains the posterior chain muscles more effectively than any other movement that uses them because none of the other movements involve enough range of motion to use them all at the same time, and none of the other movements train this long range of motion by preceding their *concentric*, or shortening, contraction with an *eccentric*, or lengthening, contraction, which produces a stretch-shortening cycle, or *stretch reflex*.

The squat's stretch-shortening cycle is important for three reasons:

1. The stretch reflex stores energy in the viscoelastic components of the muscles and fascia, and this energy gets used at the turnaround out of the bottom.
2. The stretch tells the neuromuscular system that a contraction is about to follow. This signal results in more contractile units firing more efficiently, enabling you to generate more force than would be possible without the stretch reflex.
3. Because this particular loaded stretch is provided by the lowering phase of the squat (which uses all of the muscles of the posterior chain over their full range of motion), the subsequent contraction recruits many more motor units than would be recruited in a different exercise.

The conventional deadlift, for example, uses the hamstrings and glutes, but it leaves out much of the adductors' function, and starts with a concentric contraction in which the hips start out well above the level of a deep squat. No bounce, shorter range of motion, but very hard anyway – harder, in fact, than squatting, due to the comparatively inefficient nature of the dead-stop start – yet not as useful to overall strength development. Plyometric jumps can be deep enough and might employ the requisite stretch reflex provided by the drop, but they are not incrementally increasable the way a loaded barbell exercise is, they can be damned tough on the feet and knees for novices, and they are not weight-bearing in the sense that the whole skeleton is loaded with a bar on the shoulders. In contrast, the squat uses all the posterior chain muscles, uses the full range of motion of the hips and knees, has the stretch-shortening cycle inherent in the movement, and can be performed by anybody who can sit down in a chair, because we have very light bars that can be increased in weight by very small increments.

The term "posterior chain" obviously refers to the anatomical position of these muscular components. It also indicates the nature of the problems most people experience under the bar, trying to improve their efficiency while squatting. Humans are bipedal creatures with prehensile hands and opposable thumbs, a configuration that has profoundly affected our perception as well as our posture. We are used to doing things with our hands in a position where our eyes can see them, and we are therefore set up to think about things done with our hands. We are not used to thinking about our nether regions, at least those unrelated to toilet functions. The backside of your head, torso, and legs are

seldom the focus of your attention unless they hurt, and they remain visually unobservable even with a mirror. The parts you can see in the mirror – the arms, chest, and abs, and the quads and calves if you're wearing shorts – always end up being the favorite things for most people to train. They are also the easiest parts to learn how to train because they involve or are facilitated by the use of our hands, and we are very "handsy" creatures.

The hard parts to train correctly are the ones you can't see. The posterior chain is the most important component of the musculature that directly contributes to gross movement of the body, as well as being the source of whole-body power. The posterior chain is also the hardest part to learn how to use correctly. This would be easier if you didn't have any hands: how would you pick up a table without the ability to grab the edge of the thing and lift it? You'd get under it and raise it with your upper back, or squat down and drive up with your hips against the undersurface of it, or lie down on your back and drive it up with your feet, because those would be the only options open to you. But your hands shift your focus away from these options and enable you to avoid thinking about them at all. So posterior chain matters remain largely unexplored by most people, and this makes their correct use a rather groundbreaking experience.

You will find that the posterior aspects of squatting and pulling present the most persistent problems, require the greatest amount of outside input from coaches and training partners, and will be the first aspects of form to deteriorate in the absence of outside reinforcement. For coaches, the posterior chain is the hardest part of the musculature to understand, to explain, and to influence. But it is also the most critical aspect of human movement from the perspective of athletic performance, and the mastery of its lore can determine the difference between an effective coach and a slightly-more-than-passive observer, between an effective athlete and one who merely moves.

Much is made of "core" strength, and fortunes have been made selling new ways to train the core muscles. A correct squat perfectly balances all the forces around the knees and the hips, using these muscles in exactly the way the skeletal biomechanics are designed for them to be used, over their full range of motion. The postural muscles of the lower back, the upper back, the abdominals and lateral trunk muscles, the costal (rib cage) muscles, and even the shoulders and arms are used isometrically. Their static contraction supports the trunk and transfers kinetic power from the primary force-generating muscle groups to the bar. The trunk muscles function as the transmission, while the hips and legs are

the engine.

Notice that the “core” of the body is at the center of the squat, that the muscles get smaller the farther away from the “core” they are, and that the squat trains them in exactly this priority (Figure 2-2). Balance is provided by the interaction of the postural muscles with the hips and legs, starting on the ground at the feet and proceeding up to the bar. Balance is controlled by a massive amount of central nervous system activity under the conscious direction of the athlete’s mind. In addition, the systemic nature of the movement, when done with heavy weights, produces hormonal responses that affect the entire body. So not only is “the core” strengthened, but it is strengthened in the context of a total physical and mental experience.



Figure 2-2. Total-body power development originates in the hips, and the ability to generate power diminishes with distance from the hips. Note also that the farther from the center of the body a body part is, the greater the angular velocity with which the body part can move, enabling the application of power through acceleration. From a concept by David Webster, versions of which have been used by Tommy Kono and Bill Starr. This concept has recently gained new traction under the names “core strength,” “core stability,” and “functional training.” It seems rather obvious to the author that an athlete with a 500-pound squat has a more stable “core” than that same athlete would with a 200-pound squat.

The squat is poorly understood because it involves the use of many muscles – more than most people realize – and most of the people who don’t understand it have never done it correctly themselves. This means that they can’t appreciate the true nature of the movement and the interactions of all the muscles functioning in a coordinated manner, since to truly understand a thing,

you must experience it personally. The more people who learn to squat correctly, the more people there will be who understand the squat, and then, like ripples in a pond, knowledge and strength will spread. This process starts here, with you.

Loaded Human Movement

A basic understanding of the nature of loaded human movement – the ways that the skeletal system translates the force of muscle contraction into movement as the body interacts with its environment – is essential to understanding barbell training. A few simpl e lessons, which can be learned through observing the squat, are equally applicable to all other barbell exercises. The most basic of observations is that when a barbell is loaded, the force that provides the weight of the barbell is gravity. And gravity – always, everywhere, every time – operates in a straight line perpendicular to the surface of the earth. Gravity is generated by mass. In this particular case, we are concerned with the mass of the planet, which has conveniently organized itself into the general shape of a sphere – ignoring minor surface features like mountains and valleys – under the influence of this gravity. So the surface of the earth is assumed to be horizontal for this definition; after all, a rock dropped on the side of a hill still falls in the direction we define as *down*. This fact has yet to be disputed, and the principle has risen to the status of Physical Law: there are no known examples of unimpeded objects falling in a path described as “non-vertical.” The force of gravity acting on the bar is always acting straight down in a vertical line. Therefore, the most efficient way to oppose this force is by acting on it vertically as well. So not only is a straight line the shortest distance between two points, but a straight *vertical* line is also the most efficient bar path for a barbell moving through space in a gravitational framework.

In fact, the work done on a loaded barbell must be analyzed on the basis of this framework. *Work* is defined as the amount of *force* (the influence which causes a change in motion or shape) multiplied by the *distance* the barbell moves. Pounds on the bar being a unit of force, work can be expressed in foot-pounds. But since gravity operates in only one direction, straight down, the work done against gravity consists only of the distance the barbell moves vertically. Any other motion imparted to the bar – i.e., horizontal motion, in a direction either forward or back relative to the lifter – cannot be considered work against gravity, although force will be utilized when the motion is produced. Rolling the barbell around the room constitutes work against gravity *only if the elevation of the barbell changes*, because gravity influences the mass of the barbell in only

one direction – down.

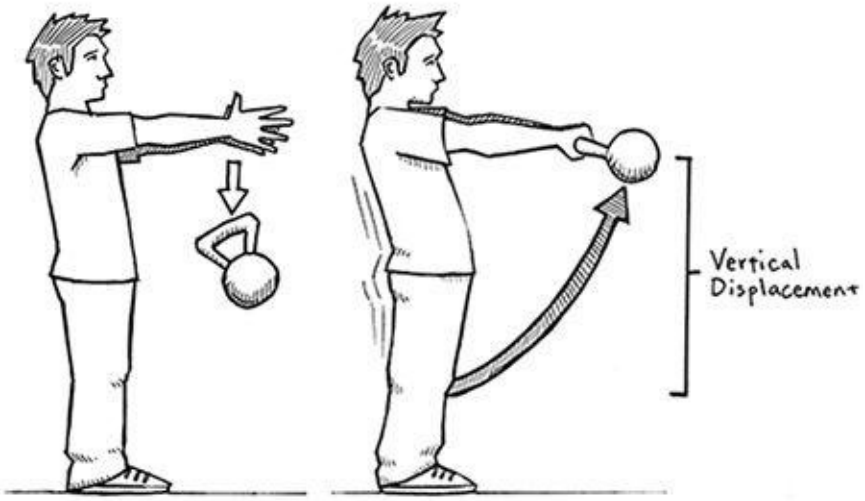


Figure 2-3. Gravity acts vertically, and only vertically. Any work done against gravity will be done in a direction opposite to its force, *i.e.* straight up. Any horizontal component to a barbell movement is not work done against gravity.

Next, when a barbell is supported by a human body, the lifter and the barbell must be considered as a system for any analysis that applies to their combined mass. The center of mass (COM) of the human body in the standing "normal anatomical position" is a point in the middle of the hips, approximately level with the sacrum. When you squat down below parallel, the geometry of the system changes to place the COM in the air somewhere between your thighs and your torso. The COM of the loaded bar is in the middle of the bar on your back. The *lifter/barbell system* has a COM somewhere between the two. As the weight of the bar increases, the system's COM moves closer to the bar until, at very heavy weights, the barbell itself approximates the system's COM. For practical purposes, we will assume that the barbell will be loaded with heavy weights and that the barbell is usually the object that we must be concerned with balancing as we move it through the range of motion of the exercise.

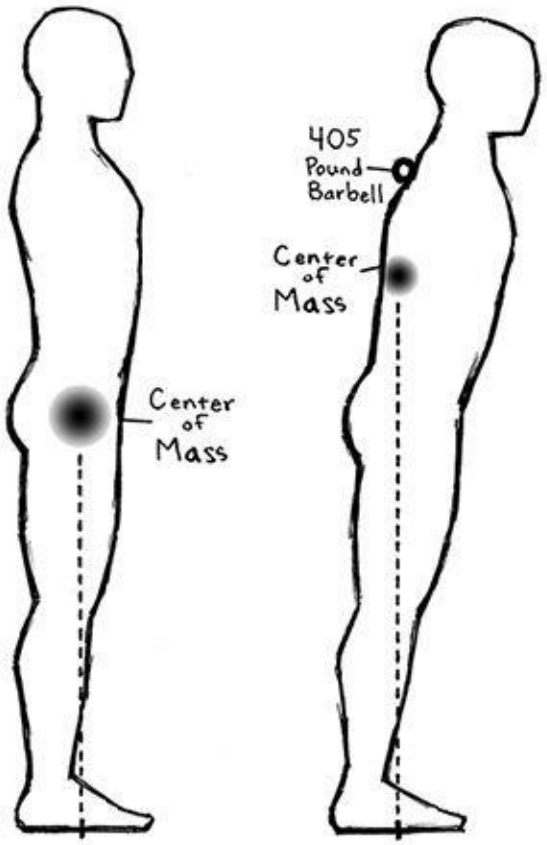


Figure 2-4. The COM shifts up toward the bar as the mass of the barbell increases.

Notice that in Figure 2-5, a dashed line illustrates a vertical relationship between the barbell on the back and the middle of the foot against the floor. It should be intuitively obvious that the lifter/barbell system will be in balance when it is directly over the middle of the foot, with the *midfoot* position – right under the arch of the foot – being the point of interaction with the ground that is the farthest away from both the forward and rearward edges of contact. Very simply, the midfoot is exactly halfway between either end of the sole of the shoe. It is therefore the most stable position, the one which would take the most movement to disrupt, and therefore the one naturally favored by the body, loaded or not. The heavier the weight on the bar, the more precisely the bar position calibrates to the midfoot. In other words, at light weights, where the mass is primarily that of the body itself, the bar may be forward of the midfoot in a position of stability, and as the weight increases, the bar comes into balance more directly over the midfoot.

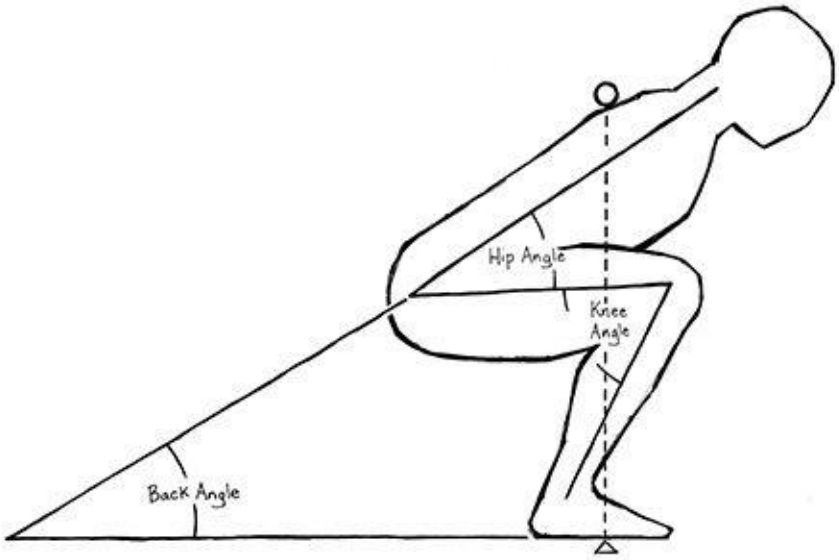


Figure 2-5. The diagnostic angles for the squat. The hip angle is formed by the plane of the torso and the femur. The knee angle is formed by the femur and the tibia. The back angle is formed by the plane of the torso and the floor. Note that the barbell is directly over the midfoot and is therefore in balance.

The body prefers stability to pretty much everything else. For example, the ankle joint – the actual point of rotation – is behind the midfoot, and the calf muscles attach at the heel at about the same distance behind the ankle as the midfoot is in front of it. The calf muscles exert tension on the heel behind the ankle to counter the effects of the leverage between the ankle and the midfoot (Figure 2-6). The body selects the midfoot as the balance point by inclining the shins and doing the calf work necessary to maintain this more stable position. In addition, the gastrocnemius, the hamstrings, and the quadriceps all cross the knee joints, stabilizing the position of the knees relative to the ankles, and the hips are embedded in a web of muscle, tendon, and ligament that permits the upright body to squat down under load and maintain a position of balance over the midfoot.



Figure 2-6. The midfoot balance point is the position favored by the body for balance. The point of rotation at the bottom of the leg – the ankle – does not function as the last piece of the kinetic chain due to the stability provided by the anchoring system of the lower leg, calf muscles, and foot; this system maintains the tibial angle and transfers force to the sole of the foot. Considering the system this way allows us to calculate balance from the midfoot position, the point of greatest stability against the floor.

Consider the unloaded lifter: if you stand up straight with your hands on your hips and lean forward, even a little, you can feel the weight shift to the balls of your feet and feel the increased tension in your calves as you apply some force to the mass of your body above your feet to keep from falling forward. If you lean back, you can feel the shift onto your heels – lean back far enough, and you will have to actually hold your arms out in front of you to change your center of mass so that you don't fall back. (Our bodies have evolved to move forward, and forward imbalances are more naturally handled by our anatomy.) You settle into a position of balance when the greatest amount of force is needed to perturb the position, or when the least amount of force is needed to maintain the position. When you stand, this position is where your COM is over the midfoot, and when you squat down and stand back up, your body's COM is in balance when it travels in a vertical line directly over this point. Since you will do most barbell exercises (except the bench press) while standing on your feet, this midfoot balance point becomes a critically important concept in the analysis of good exercise technique.

Let's assume that the bar in Figure 2-5 weighs 315 pounds. Were the bar

forward of this balance point, it would still weigh 315 pounds, but the effort required to move it through its range of motion would be greater. The eccentric and concentric work done on the 315 pounds would be harder due to the bad leverage position generated by the distance the bar was out of balance. And the isometric stress of stabilizing the load in the bad leverage position adds quite significantly to the effort. Keeping the 315-pound bar directly over the midfoot through the complete range of motion (ROM) constitutes the most efficient way the work *should* be done during the lift. When the bar is off-balance, the added energy you must expend due to the leverage of the off-balance load makes 315 much harder to lift.

It doesn't take much of an imbalance for the leverage to increase to the point where the rep is missed. Imagine the bar on your back in a position 12 inches in front of the midfoot as you try to squat; this is an awkward position with even 30% of your 1RM (1 rep max), and the heavier the weight gets, the smaller the imbalance you can deal with. You can easily see that this continuum ends up with essentially *zero* amount of deviation tolerable at 1RM loads. This concept applies to every barbell exercise where the load must be balanced. So, "good technique" in barbell training is easily and understandably defined as the ability of the lifter to keep the bar vertically aligned with the balance point. The ability to maintain this balanced relationship between the bar and the ground is one of the many things trained with barbells that are not trained in other exercise methods. Since balance is an important characteristic of most human physical endeavors, this is one more reason to base your training on barbell exercises.

Figure 2-5 also shows the angles we use to analyze the movement of the body under the bar during the squat. The *hip angle* is the angle formed by the femur and the plane of the torso. Even though the spine is curved when held in the correct position to bear weight under the bar, it is held rigid during the squat, so we can use the concept of "the plane of the torso" to describe the mechanical behavior of this segment under the bar. The *knee angle* is formed by the femur and the tibia, effectively illustrating the relationship between the thigh and the "shank" (as the lower leg segment is called). The *back angle* is formed by the plane of the torso and the floor, which is assumed to be horizontal (meaning *level*, perpendicular to the force of gravity).

These angles describe the relationships of their constituent segments to each other under the load of the barbell. The back angle is said to be either more *vertical* or more *horizontal*, while the knee and hip angles are either more *open* or more *closed*. Control of the position of these angles depends on the muscles operating the bones that form the angles. We know that the lifter/barbell system will be in balance when the bar is directly over the middle of the foot, and the

heavier the bar, the more precisely this position must be kept. Even if the weight is light enough to remain in a position of imbalance, the lifter will expend more energy than he would if the bar were in balance.

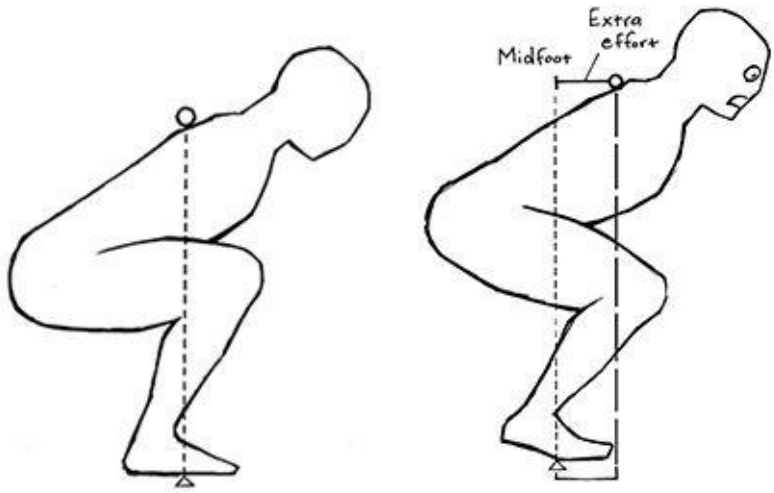


Figure 2-7. Extra work that must be done on an out-of-balance bar.

If the bar is on the front of the shoulders, as in the front squat, this bar position will require a very vertical back angle if the bar is to be kept over the midfoot, as Figure 2-8 illustrates. Notice the knee angle made necessary by this position: it is very closed. And notice the hip angle: it is much more open than it would be with a more horizontal back angle. In this position, the hamstrings are shortened because their proximal attachments on the pelvis and their distal attachments at the knee are as close together as they can be at the bottom of a squat. Here, the hamstrings are functioning isometrically to hold the torso in the nearly vertical position required of the front squat, a much easier position to hold than a more horizontal back angle because of the reduced leverage against the hips (much more on this later). But when the hamstrings are shortened, there is not enough contractile capacity left to contribute much to hip extension. In essence, the hamstrings are already contracted in the bottom of the front squat and can't contract much more. This leaves the glutes and adductors on their own to produce hip extension, and this is why your butt gets so sore when you front-squat heavy: it's having to do all the work the hamstrings normally help with in a squat.

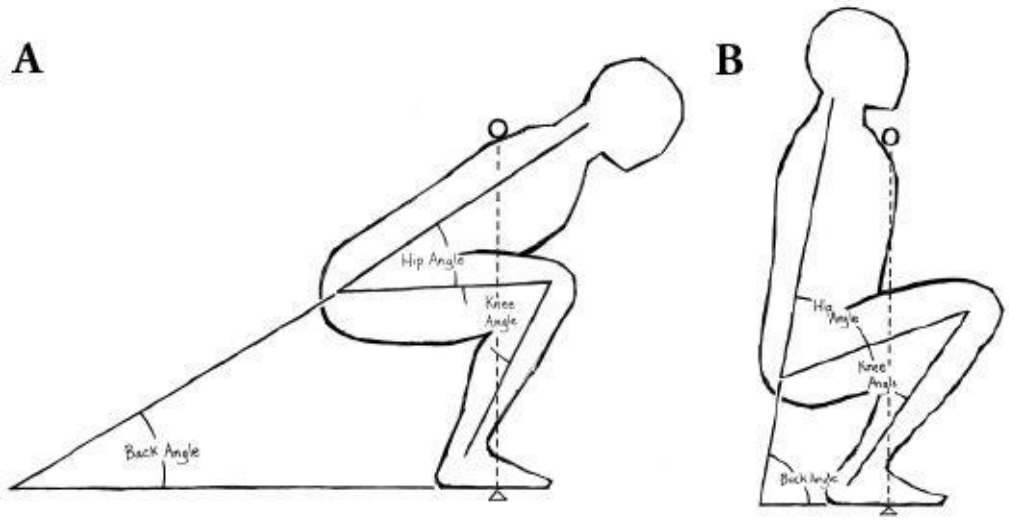


Figure 2-8. Squat variations commonly seen in the gym. (A) The low-bar squat, our preferred position and the form referred to in this text as "the squat." (B) The front squat, used to catch and recover from a clean and as an assistance exercise by Olympic weightlifters.

The upshot of this situation is that the front squat leaves out much of the hamstrings' function, and we'd like to use the hamstrings when we squat so that we can get them strong. The front squat is therefore a poor choice for training the posterior chain. To best recruit the hamstrings, and let them contribute the most they can to hip extension, we need to use a squat form that produces a more closed hip angle and a more open knee angle. At the bottom of this squat, the hamstrings are contracted isometrically – that is, they are stretched out proximally, by the attachments at the pelvis, even as they are shortened distally because of the flexing knees. As the knees and hips extend during the ascent, the hamstrings have to work hard to maintain tension on the pelvis, and to control the effects of the increased leverage demands of the more-horizontal back angle. The back angle largely determines the hip angle, and the back angle enables the hamstrings to contribute more force to the squat.

And when we use that more horizontal back angle, the bar must be placed on the back such that the bar is over the middle of the foot. The lower the bar is on the back, the more horizontal the back angle can be. The bar should therefore be in the lowest secure position it can occupy on the back, right below the spine of the scapula – that bump on your shoulder blade you can feel when you reach across and touch the back of your shoulder. Any lower than this, and the bar scoots down a little every rep of the set.

If the adductors – the groin muscles – get their share of the load, too, that

adds muscle mass to the exercise. When we use a moderate stance with shoulder-width heels, toes pointing out at about 30 degrees, and knees shoved out so that the thighs stay parallel to the feet, then the groin muscles stretch out as the hips are lowered. If the muscles are stretched out, they are in the position they must be in to contract and contribute force to the hip extension. The muscles that hold the knees out – the external rotators of the hip – are engaged as well, thus adding to the muscle mass involved in the squat.

The low-bar squat, or in this book, just the Squat, is not the same form used by suit-and-wraps-equipped powerlifters, who are trying to get the most out of their squat suit, an expensive, very tight singlet that is designed to resist hip flexion and store elastic energy in the eccentric phase, and therefore aid hip extension. To this end, some powerlifters use a very wide stance and as vertical a shin position as they can produce. Some lifters use a high-bar position with low elbows, a more vertical back angle, and an upward eye gaze (quite different from the squat style used in this book). A wide stance and vertical shins open the knee angle and close the hip angle, thus permitting the more effective use of the suit/hip extension. Knee wraps are used to resist knee flexion, and like the squat suit, they store elastic energy during the eccentric phase. Our stance, which is not nearly as wide, permits more forward travel of the knees and more use of the quadriceps. In fact, every aspect of the technique used in our version of the squat has been chosen specifically to maximize the amount of muscle mass and the range of motion used so that we can lift as much weight as possible through that range of motion and thus get stronger.

If the bar is placed high on the back – on top of the traps, where most people start off carrying it because it's an easier and more obvious place for a bar – the back angle must accommodate the higher position by becoming more vertical to keep the bar over the midfoot. If the back angle is more vertical, the knee angle must become more closed because the knees get shoved forward when the hips open up (Figure 2-8 again). In other words, the higher bar position makes the back squat more like the front squat, and we don't want to front-squat for general strength development because it doesn't effectively train the source of whole-body power: the posterior chain.

The high-bar, or "Olympic," squat has been the preferred form of the exercise for Olympic weightlifters for decades. This seems to be largely a matter of tradition and inertia, since there are compelling reasons for weightlifters to use the low-bar position, too. Since the squat is not a contested lift in weightlifting, and since Olympic lifters front-squat to directly reinforce the squat clean anyway, the reasons for weightlifters to use the low-bar squat in training must involve other considerations. The squat makes you strong, and

weightlifting is a strength sport; even if it is terribly dependent on technique, the winner is still the one who lifts the most weight. The high-bar position may be harder, but the low-bar position uses more muscle, allows more weight to be lifted, and consequently prepares the lifter for heavier weights.

If an argument on the basis of specificity is to be made, the low-bar squat is also more applicable to the mechanics of Olympic weightlifting than the high-bar squat. The low-bar position, with the weight sitting just below the spine of the scapula, much more closely approximates the mechanics of the position in which the bar is pulled off of the floor. As the discussions of pulling mechanics in the Deadlift and Power Clean chapters illustrate, the shoulder blades are directly above the bar when it leaves the floor in a heavy pull, and they stay there until the bar rises well above the knees. This is true for both the clean and the snatch, with the snatch being done from a position even less similar to the Olympic squat than the clean is. Low-bar squats done with this similar, relatively horizontal back angle train the movement pattern more directly than does the high-bar version, which places the back at a higher angle due to the higher position of the bar on the traps. And they do it through a nice, long range of motion due to the fact that the squat goes to a deeper hip position than the start position of either the snatch or the clean and jerk.

If the back angle is kept constant for both the low-bar squat and the pull from the floor (which it must be, see back angle discussion in the deadlift chapter), they are very similar movements – more similar than a high-bar squat and a pull of any type. If an argument is to be made for squatting with a form specific to the motor pathway requirements of the sport, the low-bar position would be that form. And if an argument is made that the squat need not be similar, the low-bar squat still makes more sense because it can be done with heavier weights.

Squat Depth – Safety and Importance

The full squat is the preferred lower-body exercise for safety as well as for athletic strength. The squat, *when performed correctly*, not only is the safest leg exercise for the knees, but also produces more stable knees than any other leg exercise does. Correctly is deep, with hips dropping below level with the top of the patellas (see Figure 2-1). Correctly is therefore full range of motion.

Any squat that is not deep is a partial squat, and partial squats stress the knees and the quadriceps without stressing the glutes, the adductors, or the hamstrings. In full squats, the hamstrings, groin muscles, and glutes come under

load as the knees are shoved out, the hips are pushed back, and the back assumes the correct angle on the way down for hip drive to occur on the way up. At the very bottom of the squat, the hips are in flexion and the pelvis tilts forward with the torso. In this deep squat position (Figure 2-9), several muscle groups reach a full stretch: the adductors (attached between the medial pelvis and various points on the medial femur), and the glutes and external rotators (attached between the pelvis and the lateral femur). Here, the function of the hamstring muscles (attached to the tibia and to the ischial tuberosity of the pelvis), is primarily isometric, since they don't necessarily change length on the way down. In the bottom of the squat, the tightened hamstrings, adductors, glutes, and external rotators provide a slight rebound, which will look like a "bounce"; this is the stretch reflex we discussed earlier. The tension of the stretch pulls backwards on the tibia, balancing the force produced by the anterior quadriceps attachment on the tibial tuberosity. The hamstrings finish their job, with help from the quads, adductors, and glutes, by extending the hips.

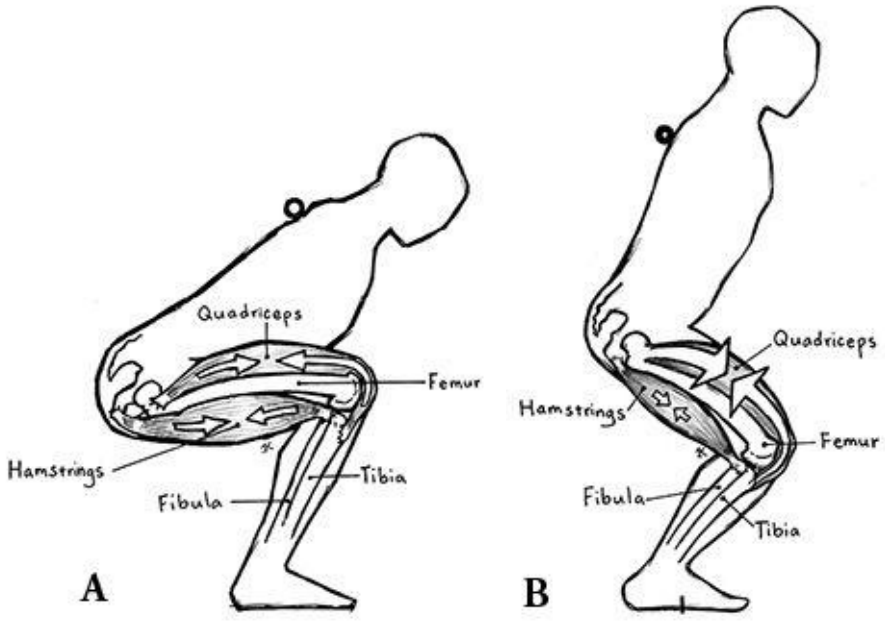


Figure 2-9. Muscular actions on the knee. In the deep squat position (A), the anterior force provided by the quadriceps is balanced by the posterior force provided by the hamstrings. The depth is the key: partial (high) squats (B) predominantly work the quadriceps and therefore lack balance.

A partial squat done with an upright torso and vertical back angle is typical of most people's attempts to squat, because we have all been told that the back must be vertical to reduce *shear*, the sliding forces that occur along a

segment in rotation. Shear between the vertebral segments is supposed to somehow disarticulate your spine, despite the fact that this cannot and has not ever occurred. But in a misinformed effort to protect the back, this advice results in a lot of unnecessary stress on the knees. As we've already discussed, however, the vertical back angle fails to fully load the hamstrings. Therefore, they cannot exert the posterior force needed to oppose and balance the anterior force exerted by the quadriceps and their attachments to the front of the tibia, below the knee. (In other words, there's no force pulling backwards to balance the forces that are pulling the knees and tibias forward.) The result is an actual anterior shear on the knee. And like a front squat, the partial squat also forces the knees quite forward of the midfoot – much more so than the low-bar squat form we will be using, which keeps the knees back and uses the hips as the primary mover of the load. This lack of posterior support produces an anterior-dominant force distribution on the knee: the further back the hips are, the more hip muscle you use, and the further forward the knees, the more quad you use. Many cases of patellar tendinitis have been caused by this incorrect squat technique. Even when partial squats are done with the correct back angle, they fail to work the full range of motion and therefore fail to perform to their potential as an exercise.



Figure 2-10. The variation in squat depths commonly seen in the gym. *Left to right:* Quarter-squat, Half-squat, a position often confused with parallel, where the undersurface of the thigh is parallel with the ground, Parallel squat according to the criteria established in Figure 2-1, and "Ass-to-grass" squat.

The hamstrings benefit from their involvement in the full squat by getting strong in direct proportion to their anatomically proper share of the load in the movement, as determined by the mechanics of the movement itself. This fact is often overlooked when the medical community considers anterior cruciate ligament (ACL) tears and their relationship to conditioning programs. The ACL stabilizes the knee: it prevents the tibia from sliding forward relative to the

femur. As we have already seen, so does the hamstring group of muscles. Underdeveloped, weak hamstrings thus play a role in ACL injuries, and full squats strengthen the hamstrings. In the same way the engaged hamstrings protect the knees during a full squat, hamstrings that are stronger due to full squats can protect the ACLs during the activities that we are squatting to condition for. With strong hamstrings and the knees-back position provided by the low-bar version of the squat, the hips bear most of the stress of the movement. So athletes who are missing an ACL can safely squat heavy weights because the ACL is under no stress in a correctly performed full squat (see Figure 2-11).

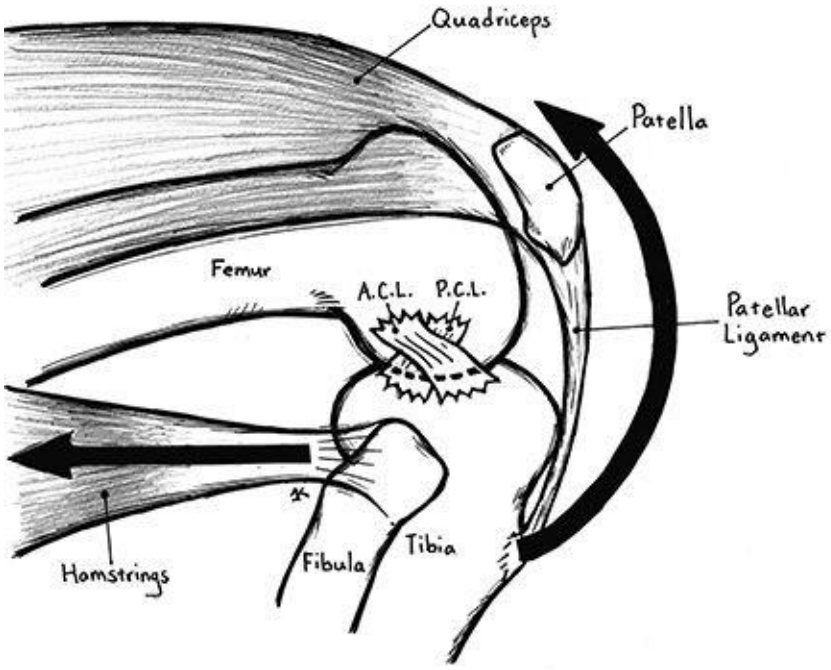


Figure 2-11. Forces on the knee in the squat. The hamstrings and adductors exert a posterior tension on the tibia, and the net effect of the anterior quadriceps tendon insertion is an anterior force against the tibial plateau. With sufficient depth and correct knee position, anterior and posterior forces on the knee are balanced. The anterior (ACL) and posterior cruciate ligaments (PCL)stabilize the anterior and posterior movements of the distal femur relative to the proximal tibia. In a correctly performed squat, these ligaments are essentially unloaded.

Another problem with partial squats is the fact that very heavy loads can be moved due to the short range of motion and the greater mechanical efficiency of the quarter-squat position. A trainee doing quarter-squats is predisposed to back injuries as a result of the extreme spinal loading that comes from putting a

weight on his back that might be more than three times the weight that he can safely handle in a correct deep squat. A lot of football coaches are fond of partial squats because they allow the coaches to claim that their 17-year-old linemen are all “squatting” 600 pounds. Your interest is in getting strong (at least it should be), not in playing meaningless games with numbers. If it’s too heavy to squat below parallel, it’s too heavy to have on your back.

There is simply no other exercise, and certainly no machine, that produces the level of central nervous system activity, improved balance and coordination, skeletal loading and bone density enhancement, muscular stimulation and growth, connective tissue stress and strength, psychological demand and toughness, and overall systemic conditioning than the correctly performed full squat. In the absence of an injury that prevents its being performed, everyone who lifts weights should learn to squat, correctly.

Learning to Squat

We will approach the squat in two phases: first unloaded, to solve problems associated with the bottom position, and then loaded, to learn how to apply the bottom position to the hip drive used for heavier weights. Since the majority of the problems with the squat happen at the bottom, this method expedites the process quite effectively.

Generating hip drive

We will use a fairly neutral foot placement, with the heels about shoulder width apart and the toes pointed out at about 30 degrees. An excessively wide stance causes the adductors to reach the end of their extensibility early, and excessive narrowness causes the thighs to jam against the belly. Both of these problems prevent you from reaching proper depth. Shoulder width is proportionate to pelvic width in most people, and experience has shown that this width works well for most of the population. **Many people will assume a stance with toes pointed too forward, so you may need to point them out more than you want to.** Look down at your feet and make a mental picture of what you see.

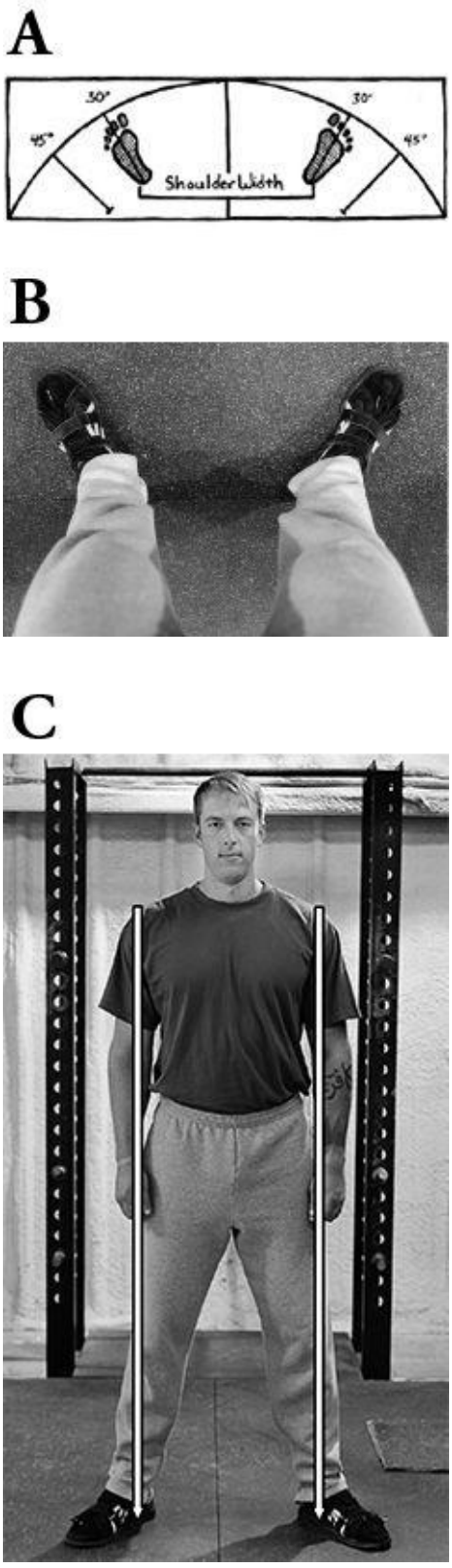


Figure 2-12. (A) Map of foot placement and (B) stance in the squat, as seen from above. (C) Heel

placement by shoulder width.

Now comes the crucial part of learning the movement. You are going to assume the position you will be in at the bottom of a correct squat, without the bar. This method works well because you can easily correct any errors in position before the bar adds another variable to the system. And if you've already been in the correct bottom position *without* the bar, getting into that position again *with* the bar is easy. Assume the correct stance and squat down, all the way. Don't even think about stopping high; just go on down to the bottom. Sometimes a lack of flexibility or a failure to point your toes out enough will alter your stance on the way down, so make sure you have assumed the correct foot position.

Next, put your elbows against your knees, with the palms of your hands together, and shove your knees out (Figure 2-13). This will usually be a decent bottom position, and if your flexibility is not great, the position will act as a stretch if you maintain it for a few seconds. Remember, **proper depth is essential in the squat**, and this low bottom position lays the groundwork for your attaining good depth from now on.



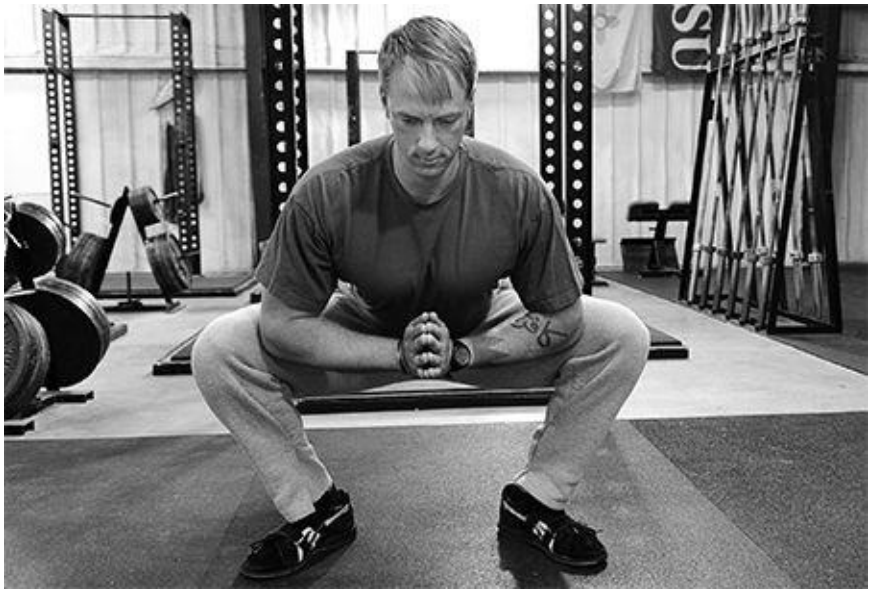


Figure 2-13. Use your elbows to stretch into the correct position at the bottom. The femurs are parallel to the feet, the feet are flat on the ground at the correct angle, the hips are back, the knees are just a little forward of the toes, and the back is at an angle (about 45 degrees) that will place the bar over the middle of the foot.

Stay in the bottom position for a few seconds to allow for some stretching. If you get fatigued by holding the position, your flexibility might not be quite what it should be. Stand up and rest for a few seconds. Then go back down to get some more stretching done and to reinforce your familiarity with the bottom position. This is the most important part of learning to squat correctly because good depth is the difference between a squat and a partial squat.

Now is the time to notice some important details about the bottom position. Your feet are flat on the floor, your knees are shoved out to where they are in a parallel line with your feet, and your knees are just a little in front of your toes. Your back should be as flat as you can get it, but if it's not perfect, we'll fix it later. Also notice that your back is inclined at about a 45-degree angle, not at all vertical. You may think it's vertical, but it won't be and it's not supposed to be. And your eyes are looking down at the floor a few feet in front of you.

After you've established the bottom position, come up out of the bottom by driving your butt straight up in the air. *Up*, not forward. This movement keeps your weight solidly over the whole foot instead of shifting it to the toes. Think about a chain hooked to your hips, pulling you straight up out of the bottom (Figure 2-14). Don't think about your knees straightening out, don't

think about your feet pushing against the floor, and don't even think about your legs. Just drive your hips up out of the bottom, and the rest will take care of itself.



Figure 2-14. An interesting way to visualize hip drive in the squat.

This important point should not be missed. Our previous discussion about hip drive and the use of the hamstrings in the squat applies here. The squat is not a leg press, and the idea of pushing the floor with the feet provides an inadequate signal for the hamstrings, adductors, and glutes to provide their power out of the bottom. Hip extension is the first part of the upward drive out of the bottom. When you think about raising your butt up out of the bottom, the nervous system has a simple, efficient way to fire the correct motor units to initiate hip drive.

Eye gaze direction plays an important part in this process of driving the hips, and it is introduced even before the bar becomes part of the squat. Looking up at the ceiling when squatting has so many detrimental effects on proper technique that it is absolutely amazing that so many people still advise their lifters to do it. It interferes with the correct bottom position, with hip drive out of the bottom, and with correct chest position. It changes the focal point from a close, manageable spot to one that is farther away. And the neck position that results from looking at the ceiling is inherently unsafe: to place the cervical spine in extreme overextension and then to place a heavy weight on the trapezius muscles directly underneath it is, at best, imprudent. The normal anatomical

position for the cervical spine is the preferred position when the weights get heavy.

The habit of looking up is also a very difficult problem to correct if it has existed for any length of time. Lifters whose high school football coaches taught them to look up during the squat often have a very difficult time with changing the eye gaze direction, even when we have effectively demonstrated that looking down works so much better. An embedded movement pattern is always easier to perform than a new one, and it will be the default movement pattern if conscious control is shifted to another aspect of the new technique.

Do an experiment or two to demonstrate for yourself the effect of gaze direction. Assume the bottom position with knees out, toes out, and heels down. Put your chin down slightly and look at a point on the floor 4 or 5 feet in front of you. Now drive your hips up out of the bottom, and take note of how this feels. Now do the same thing while looking at the ceiling. If you have a training partner or coach, get in the bottom position and have him block your hips, with a hand placed firmly on your lower back and pushing straight down, so that you have something to push up on, but not so that he pushes you forward. Push up against the resistance while looking down at your floor focus point, and note the effectiveness of your hip drive and the power it produces. Then try this movement again while looking up. You will discover an amazing thing – that the chin-down (looking down keeps the chin down), eyes-down position enables your hip drive to function almost automatically. In contrast, the upward eye gaze pulls the chest forward, the knees forward, and the hips forward – just a little, but enough to produce a profound effect. It slacks the hamstrings and all the posterior muscles we are trying to keep tight so that we can use them to drive the hips up. The first time you do this experiment will convince you that looking down is more efficient.



Figure 2-15. Blocking the hips to learn the effect of eye gaze direction. An upward-directed gaze quite effectively diminishes the ability to use the posterior chain during the drive up from the bottom.

Looking at the floor also provides the eyes with a fixed position reference. Using this reference, you can easily identify any deviation from the correct movement pattern and adjust it as it happens. The ceiling also provides a reference, but the neck position is unsafe, and anything you're looking at upward will be farther away than the floor when you're at the bottom of the squat. It's hard to imagine a room in which the floor isn't closer to the eyes than the ceiling is; the floor is therefore more useful as a reference – smaller movements can be detected against the closer point.

Most people will have more trouble with this change in their eye gaze direction than with any other aspect of this squatting method. To correct the error of looking up, fix your eyes on a position on the floor 4 or 5 feet in front of you. If you're training close to a wall, find a place to look at that is low on the wall and results in the same neck position. Stare at this point, and get used to looking at it so that it requires no conscious effort. Most people, if they are looking down, will not raise their heads to the point where neck position is affected. Inventive coaches have used tennis balls for the purpose of demonstrating a chin-down, chest-up position (Figure 2-16).

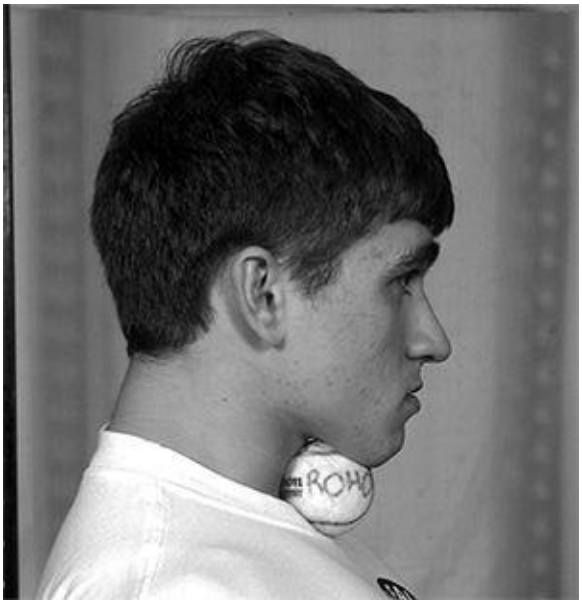


Figure 2-16. A tennis ball can teach the correct chin/neck relationship.

Adding the bar

Now you're ready to squat. You have already been in the position you will go to at the bottom, and now you're just going back down there with the bar. First, chalk your hands. Chalk is always a good idea because it dries out the skin. Dry skin is less prone to folding and abrasion than moist skin and therefore is less prone to problem callus formation. If the weight room is not equipped with chalk, bring your own. If the gym complains, change gyms.

The squat begins at the power rack or the squat stands, whichever is available. Set the rack height so that the bar in the rack is at about the level of

your mid-sternum. Many people will perceive this as too low, but it's better to be a little low taking the bar out of the rack than to have to tiptoe back into the rack with a heavy weight. Often, this position in the empty rack will look low because the diameter of the bar sitting in the hooks tells the eye a different story about its true height in the rack. When the bar is placed in the rack, the eye will be more comfortable with the setting. And remember, we are placing the bar in a lower position than the top of the traps, so you'll need the rack lower than you think. You'd rather have the rack set a little too low than a little too high, and most people are not as tall as they think they are. *Most people will want to use a position in the rack that is too high.* If your shoulders are not flexible enough to assume the low-bar position at first, they should stretch out over a couple of weeks.

Face the bar. Always an empty bar at first. ALWAYS. There will be plenty of time very soon to add weight. Take an even grip on the bar, measured from the markings placed on the bar for this purpose. A standard power bar has 16–17 inches between the ends of the outside knurl, and 32 inches between the finger marks, those one-eighth-inch gaps in the knurl indicating a legal bench-press grip. Grip width for the squat will obviously vary with shoulder width and flexibility, but in general, the hands will be between these two markings on this type of bar. A narrower grip allows a flexible person to better support the bar with the posterior muscles of the shoulders when the elbows are lifted, and a wider grip allows an inflexible person to get more comfortable under the bar. In either case, a narrower grip tightens your shoulder muscles so that the bar is supported by muscle and doesn't dig into your back.



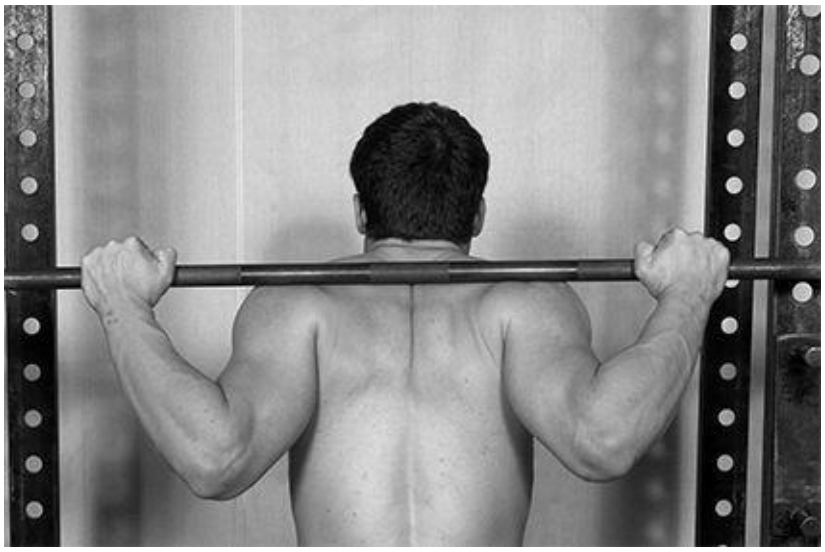


Figure 2-17. A comparison of wide and narrow grips. Note the difference in tightness of the upper back muscles and the resulting difference in bar support potential.

The thumbs should be placed on top of the bar so that the wrists can be held in a straight line with the forearms. The elbows should be cranked up to trap the bar between the hands and the back. If a lack of flexibility in the chest and shoulders prevents your achieving this position, use the high-bar position until proper stretching can make you flexible enough to get the bar down to a better position. If you're flexible enough now, take a grip wide enough to permit straight wrists under the bar, and then with each set, narrow your grip a little until it is tight and secure. Mark this position as the grip you will use.





Figure 2-18. Wrist alignment on the bar. The correct grip keeps the hand above the bar and keeps all of the weight of the bar on the back. An incorrect grip intercepts some of the weight, loading the wrists and elbows. Note that the thumb is on top of the bar and the hand is between the outer ring and the inner edge of the knurling.

With your grip in place, and your hands and thumbs on top of the bar, dip your head under the bar, and come up into position with the bar on your back. Place the bar in the correct position, just immediately under the bone you feel at the top of the shoulder blades – the spine of the scapula – and then secure it in place by lifting your elbows and chest at the same time (Figure 2-20). It should feel as though the bar is resting on a "shelf" under the traps and on top of the posterior deltoids. This action tightens the muscles of your back and lifts your chest, placing the thoracic spine in an extended, straight position and thereby fixing many of the problems encountered with a round-back position. Enormous weights can be safely handled this way later. *Most people starting with this method will place the bar too high on the back*, perhaps just above the scapular spine instead of just below it. Check to make sure the bar is in the right position.

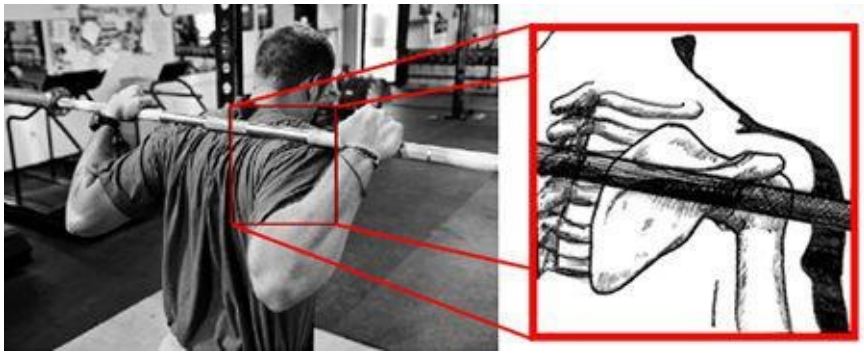


Figure 2-19. Position of the barbell relative to the scapular anatomy. The bar is just under the spine of the scapula.

First and foremost, **ALWAYS STEP BACK OUT OF THE RACK. ALWAYS. NEVER PUT THE BAR BACK IN THE RACK BY STEPPING BACKWARDS. NEVER.** This cannot be done safely. You should never be in a position to have to step backwards and rack a weight at the end of a set. You cannot see the hooks, and even if you have spotters, there will eventually be a wreck. If you do this, or permit it to be done by someone you're training, you are a fool.

Take the bar out of the rack in the same position in which it is to be squatted, with the torso and shoulders tight, the chest and elbows up, the head position down, and both feet under the bar. Everything should be the same as it is for the full squat, so take the bar off of the hooks by extending the knees and hips, just as in the top of a squat. In this way, any weight can be taken safely out of the rack. Many problems are caused by doing this improperly. It is very common to take the bar out of the rack with a loose back and chest, and then attempt to tighten everything just before squatting. It is obviously much more effective to tighten the muscles and *then* take the weight onto tight muscles than it is to take the weight, let it mash down into your back through loose muscles until it stops on some crucial skeletal component, and then try to tighten everything up underneath it.



Figure 2-20. Simultaneous lifting of the elbows and the chest "trap" the bar between the hands and the back, creating a stable back and chest position and a tight bar placement on top of the posterior delts.

Likewise, taking the bar out with one foot back and only one foot under the bar, like a lunge, is a bad habit, one that everybody gets away with when the weight is light but that can cause back problems from the unevenly stressed hips when the weight gets heavier. Unrack the bar exactly like it is in a squat, even when it is light, and you'll have no problems later when it is heavy.

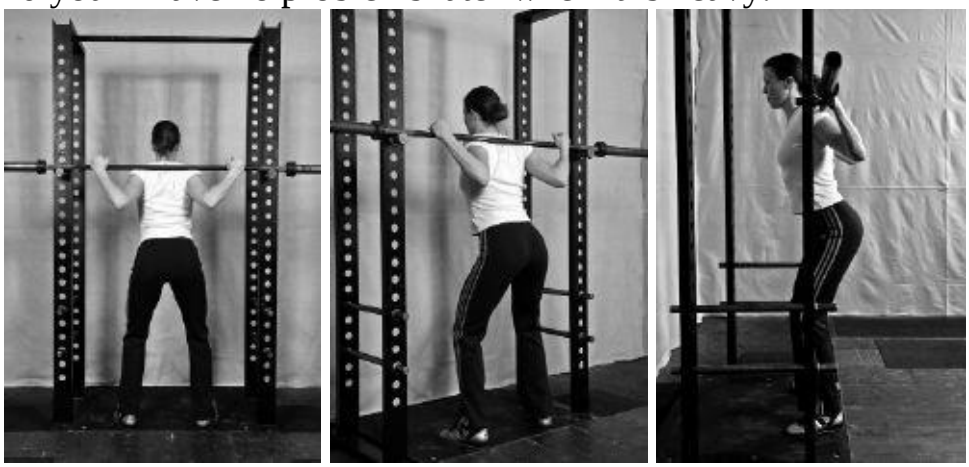


Figure 2-21. The proper position in which to receive the bar from the rack.

Once the bar leaves the rack, don't take a hike with it, backing up three or four steps before setting up to squat. This is unnecessary, and it could become a problem if the set is heavy, the spotters are unreliable, or the trip back to the rack is just too far on this particular day. One step back out of the rack with good

form is enough to clear the rack and allow the spotters to do their job while minimizing the trouble of getting the bar back home.

The stance should be the same as the one used during the stretch. Again, heels should be about shoulder width apart, with toes pointed out about 30 degrees. *Most people will change the stance at this point, rotating the toes back in.* Make sure you are using the same stance you previously used during the unweighted part of this teaching method.

At this point, you are ready to squat with the empty bar. THE EMPTY BAR. All of the groundwork has been laid, the correct bottom position is fresh in your mind, and you are now in the correct starting position. Everything you are about to do is the same as you did during the stretch. Only two things are different: one, you don't have your elbows available to help push your knees out, so you need to do this with your brain. And two, don't stop at the bottom. Just go down and immediately come back up, driving your butt straight up, not forward, out of the bottom. Now, take a big breath and hold it, look down at a spot on the floor about 4–5 feet in front of you, and squat.

You should be in good balance at the bottom of the squat, having already been there when you stretched. Your weight should stay evenly balanced over the middle of your feet.

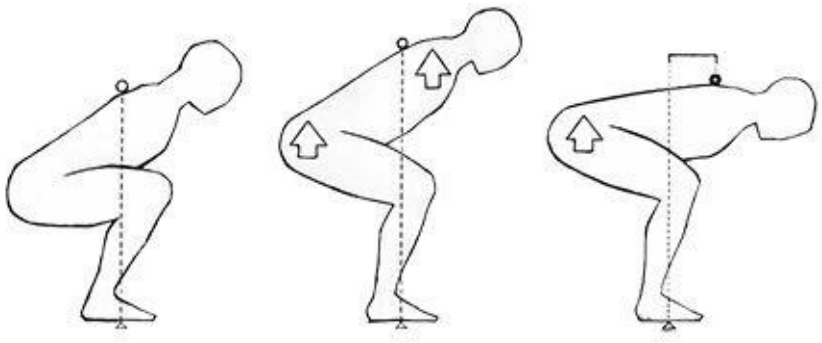


Figure 2-22. The back angle during the drive up from the bottom is critical to the correct use of the hips. The correct angle is produced when the bar is just below the spine of the scapula and directly over the middle of the foot, the back is held tight in lumbar and thoracic extension, the knees are parallel to the correctly placed feet, and the correct depth is reached. Flopping forward allows the bar to drop forward of the midfoot.

The reference point your eyes have on the floor should help you maintain position all the way down and all the way up. Balance problems usually indicate a back angle that is too vertical, so make sure you're sitting back and leaning

forward enough. *Most people have a picture in their minds of a vertical torso during the squat.* Remember that the back angle will not be vertical at all; sit back, lean forward, and shove your knees out.

Get someone to verify that your depth is good, and DO NOT accept anything less than full depth, ever, from this point on. If your impartial critic tells you that you're high, check your stance to make sure that it's wide enough but not too wide, that your toes are out enough, and that your knees are tracking parallel to your feet. While he's being helpful, get him to check your eye gaze direction and to remind you to look down every rep. If you're sure the form is fairly good, do a set of five and rack the bar. If the form is good except for the depth, the squat itself will act as a stretch IF YOUR KNEES ARE OUT. And most of the time, if you are high, it is because your knees are not out. *Most people who have problems with the squat – at this rank novice level as well as later on – do not shove their knees out enough.* If the squat is crazy bad, rack the bar and repeat the pre-squat procedure, focusing on the knees-out part.

To rack the bar safely and easily, walk forward until it touches the vertical parts of the rack. Find the uprights, not the hooks. You can't miss the uprights, and if you touch them, you'll be over the hooks. If you try to set the bar directly down on the hooks, you can and will eventually miss it on one side. Big wreck.

The general plan is to do a couple more sets of five reps with the empty bar to nail down the form, and then add weight, do another set of five, and keep increasing in even increments until the next increase would compromise the form. Sets of five are a good number to learn with – not so many that fatigue affects form during the last reps, but enough to establish and practice the technique while handling enough weight to get strong. Increments for increasing the weight between sets will vary with the trainee. Lightweight, unconditioned kids need to go up in 10–15 lb or 5–7.5 kg jumps. Older or stronger trainees can use 20–30 lb or 10–15 kg increments. Decide which jumps best fit your situation, being conservative since it is your first day. *Most people will try to increase the weight by increments that are too large for this point in the teaching method.* Go on up in weight, practicing good form and making sure to keep good depth, until you can tell that the next jump up would alter your form. Then do two more sets at the current weight, for a total of three sets across with the heaviest weight. And that is the first squat workout.

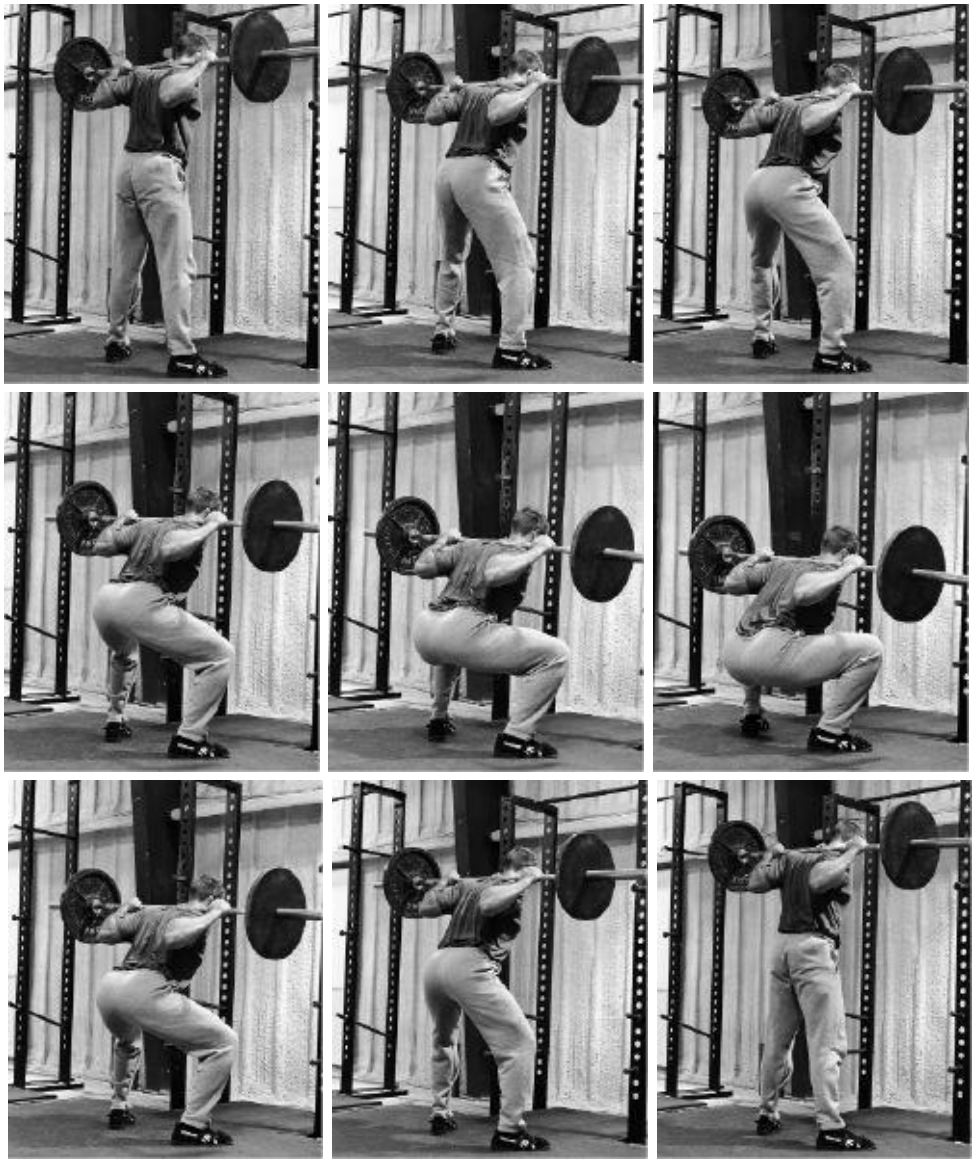


Figure 2-23. The squat.

The Important Things You're Going to Do Wrong

Depth: You're probably going to squat to a position above parallel. This will occur because you're not looking down, you're not shoving your knees out, you have a stance that is either too narrow or too wide, or you have not committed to going deep.

Knee position: You will fail to shove your knees out as you start down. This will make correct depth hard to attain and will kill your hip drive.

Stance: Your stance will be either too narrow or too wide, with your toes usually pointed too forward. This will result in a squat that is not below parallel.

Eye gaze: You will fail to look down. This will kill your hip drive.

Back angle: Your back will (usually) be too vertical, due to a faulty mental picture of what your hips do when you squat or due to the incorrect placement of the bar on your back, or your back will be too horizontal, due to your failure to keep your chest up. Either error will adversely affect hip drive and depth.

Hip drive: You will lift your chest instead of driving your hips up. This will kill your power out of the bottom by making your back angle too vertical.

Bar placement: You will place the bar too high on your back. This will adversely affect your back angle and your hip drive.

Rack height: You will set the bar in the rack in a position that is too high. This will make the preferred position on the back difficult to attain.

Notice that all of these problems are extremely interrelated. The squat is a complex, multi-joint exercise whose correct execution depends on all the components of the entire system functioning together. An incorrect placement of any component will perturb the entire system to its detriment. A working knowledge of the functional mechanics of the system is important if you are to understand the contribution of each component to the system, and the workings of the system as a whole.